KINDY-MATE: MAKE SCREEN TIME HEALTHY

> We transform children’s screen time into safe, guided, and development-focused digital experiences.



MEMBERS:

1. **Bui Vu Nhat Minh** - Leader + App designer
2. **Nguyen Nhat Minh** - App designer
3. **Cong Nghia Hoai An** - Researcher
4. **Chu Ngoc Linh** - Researcher
5. **Ha Quang Minh** - Researcher





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- Risks and Safeguards

OVER THE YEARS, SCREEN TIME HAS BECOME AN INSEPARABLE PART OF CHILDREN'S CHILDHOOD...



AVERAGE DAILY SCREEN TIME (AGES 0-8)

- Under 2: **1 hour 3 minutes/day**
- Ages 2-4: **2 hours 8 minutes/day**
- Ages 5-8: **3 hours 28 minutes/day**

Health guidelines

- Under 2: **no screen use**
- Ages 2-5: **maximum 1 hour/day**
- Ages >=6: **maximum 2 hours/day**

DEVICE OWNERSHIP

- **51%** of children (0-8) own their own **mobile device**
- **47%** own **tablets**
- **9%** own **smartphones**



OTHER CONCERNING TRENDS

- **48%** of children (0-8) have watched **short videos** on TikTok, YouTube Shorts, or Instagram Reels
- **1 in 5** families use mobile devices to manage bedtime routines, mealtimes, and emotional regulation



Children's digital exposure is no longer **occasional**. It is becoming **embedded** in **daily family routines**.

THROUGH EVIDENCES BASED ON OBSERVATIONS ALONE....



It is not difficult to see children as young as 3 years old using phones during leisure time, giving birth to unofficial terms such as "**iPad kids**"

Given parental stress and the omnipresence of smart devices
=> we suspect that screen time among children will continue to increase in near future.





Prolonged exposure to screen time causes numerous ***health problems***

EFFECTS OF PROLONGED SCREEN TIME

Universal principle: Excessive screen time is harmful not only because the content itself is bad, but because of what it **displaces -** a phenomenon called "**The displacement hypothesis**"

1

Less Physical Activity:

- Children with 2+ hours/day show more screen addiction, attention deficits, and more sedentary behavior
- Regardless of the recommended audience, screen use can replace time spent on physical activity

→ Poorer sensory processing skills
→ Predictor of lower physical wellbeing and lower autonomy
→ Lower fitness levels and energy

2

Disrupted Sleep:

- Blue light from screens significantly suppresses melatonin, the hormone that promotes sleep
- For every hour of television watched, children lose 5-10 minutes of sleep
- Mobile devices at bedtime shorten sleep by 21-45 minutes

→ Lower brain development, memory consolidation, emotional regulation
→ Affects learning capacity the next day
→ Cut off screens at least 1 hour before bedtime to protect sleep patterns

3

Reduced Real-World Interaction: Media engagement replaces critical activities for mental development:

- Parent-child interactions (crucial for attachment, language, emotional development)
- Face-to-face social engagement (builds emotional intelligence and social skills)
- Unstructured play (develops creativity and problem-solving)

→ Cannot be replicated digitally
→ Worse Language acquisition, Emotion regulation skills, Physical coordination, Social competence

HARMFUL SCREEN TIME

The most alarming issue is not **how long** children's screen time is, but **what types content** they view.

why children are more vulnerable

1. Biological Immaturity

- Developing executive functions
- Limited inhibitory control
- Underdeveloped ability to resist impulsive reactions

2. The Attention Conflict

- Dominated by "bottom-up" attention (driven by fast scenes/noises)
- Disrupted development of "top-down" control (driven by internal goals)

3. Proven Causation

- High screen time directly drives low self-control & concentration (not vice versa)

which types





CoComelon

TikTok

1. Overstimulating Videos

Compulsive autoplay
Fast-paced editing
Colorful bright colors
Little-to-no meaningful plot

2. Passive Scrolling

Infinite scroll
("drug delivery mechanism")
Low cognitive effort
One-directional experience

3. Excessive Entertainment

Age-inappropriate / Non-child-directed content
Non-educational media

4. Addictive Games

Variable rewards ("slot machine effect")
Gamification loops (streaks/badges)







HARMFUL SCREEN TIME

effects on child development

1. Shorter Attention Span

- 4% increase in inattentive behavior / hour
- Immediate damage to executive function
- Impaired memory and cognitive engagement

2. Poor Sleep

- Every hour of television, lose 5-10 minutes of sleep
- Mobile devices at bedtime can shorten sleep by 21-45 minutes

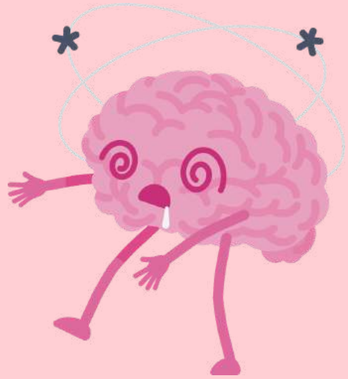
4. Weaker Social Skills

- Increased behavioral /
- aggressive problems
- Delayed language development
- Reduced quality / quantity of caregiver-child interaction

3. Emotional Dependency

- Lower emotion understanding
- Reliance on external digital regulation
- Failure to develop internal coping skills

neurobiological effects



1. Brain Structure Changes

Heightened sensitivity to immediate gratification
Continuous stimulation of the pleasure/reward circuit (VTA)

2. Algorithmic "Closed Loop"

Instant psychological gratification
Zero active search required
Erosion of independent choice

3. Brain Function Disruption

Inhibited prefrontal cortex (loss of cognitive control)
Activated default mode network (DMN) - linked to conditions like depression, anxiety, and ADHD.

HOW DO PARENTS FEEL?

Parents usually feel stressed regarding their children's screen time:

Emotional Burden & Stress Cycle

- Excessive screen time (both educational and recreational) has become a constant source of parental stress, exacerbated by the omnipresence of screens in daily life
- Parents frequently use digital devices to calm or distract children
- Children with lower self-regulation are significantly more likely to be given screens to help them relax

→ **The vicious cycle:** Parental stress → increased device use → child behavior challenges → more parental stress

Guilt, Regret & Internal Conflict

- Caregivers, particularly mothers, who alternate between permitting and limiting smartphone use due to perceived threats, often experience guilt and regret
- This inconsistent approach can send mixed signals to children and hinder the development of self-discipline in media use



Behavioral Challenges When Setting Limits

- Restrictions often provoke aggressive behavior, tantrums, and technology-related dysregulation resembling withdrawal symptoms
- Caregivers who perceived their child as behaviorally difficult were more likely to use screens as a tool for behavior regulation
- Caregivers experiencing higher levels of general distress were less likely to enforce screen time limits

Although parents aim to limit their child's phone usage, the reality often differs from intention

BENEFICIAL SCREEN TIME

However, research has shown that **not all content is harmful,** some are even **beneficial for children's development.**
If they can be leveraged, children can **develop cognitively and mentally** during leisure time.

types:









Direct audience interaction (characters talking to the child)

"scaffolding" skills

Educational Programming

Lower risk of mental health issues

Participation, decision-making, and creation

Interactive Learning

Proven to be more beneficial than passive viewing (watch-only)

Games prompting critical thinking

Creative/Problem-Solving

Storytelling-based

Encouraging movement and speech

evidence-based benefits:



Boosts Creativity & Thinking

Active participation develops executive function

Ignites Curiosity

Encourages participation in real-world/home activities

High-quality programs = Better academic performance

Accelerates Language

Joint viewing creates "richer language" density

BENEFICIAL SCREEN TIME



design principles for healthy content

Meaningful Age-appropriate developmental goals

Non-Addictive

No infinite scroll,
No autoplay,
No variable rewards

Active Participation over observation

Joyful Maintains a positive, constructive emotional tone

"Guided Play" Framework A balance between structure and child agency (freedom to choose)

CURRENT SOLUTIONS ARE NOT ENOUGH

There have been attempts to **limit children's screen time** or to **leverage educational content.** However, they are usually **flawed fundamentally**:

1. Pure Ban

- Banning screens triggers tantrums, aggressive behaviour, and withdrawal-like symptoms in children
- "No screens" is not a realistic solution for modern families who rely on devices for daily routines

2. Setting time limits

- Control screen duration but have no influence over content quality
- A child can spend their entire "allowed" time on fast-paced, overstimulating content
- Address the quantity problem while leaving the quality problem entirely unsolved

3. Kids Content Platforms (YouTube Kids, Netflix Kids)

- Business model is still attention-based: prioritise engagement over development
- YouTube Kids still exposes children to fast-paced, low-quality content despite its curated promise
- Designed around passive viewing

4. Learning Apps (Khan Academy Kids, Lingokids)

- Strong on educational outcomes but do not address children's daily entertainment needs
- Children disengage when the app feels like school rather than play
- Khan Academy Kids noted to lack guidance for children navigating its content library independently

> Families do not only need screen-time limits. They need a healthier digital environment, where the content, the structure, and the design all work for the child, not against them.

THEREFORE, OUR MISSION IS:

To create a **convenient mobile app** that can **control children’s screen time** and **transforms children's screen time from passive consumption** into **active, healthy development**.

Target audience

- **Two users:** Children aged 3–5 and 6–8, due to different screen-time-limit recommendations and different behaviors.
- **One buyer:** Aware but overwhelmed parents who know screens are a problem but need a solution that fits real family life — not just advice to use them less.

Introducing:
KINDY-MATE

AI COMPANION FOR HEALTHY SCREEN TIME



ABOUT: KINDY-MATE

Introducing the main features



FROM A CHILD'S PERSPECTIVE

Kindy-mate has 2 core features:

Limited screen time for each usage based on guidelines, requiring passwords set by parents to **lengthen the usage time**:



Kindy-Mate will accompany children across all activities

Cognitively healthy leisure activities backed by scientific evidence, divided into **5 distint sections**:



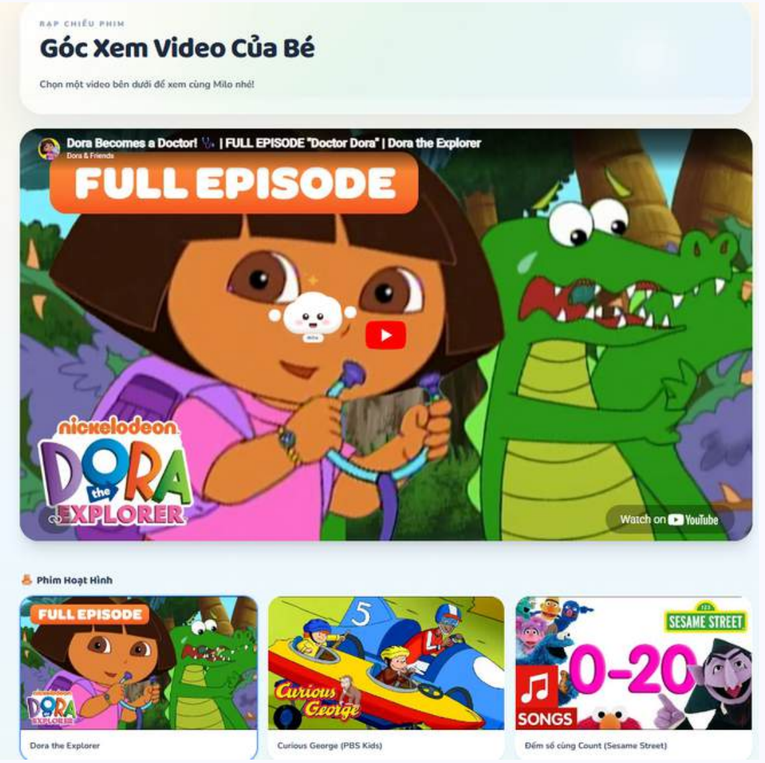
SECTION:

HEALTHY CARTOON HUB

Every cartoon on Kindy-Mate is **slow-paced, plot-driven, and prosocial-selected against 8 research-backed criteria**, not algorithmic popularity.

Content Criteria:

1. **Slow-paced** — No rapid cuts, flashing lights, or sudden loud sounds
2. **Calm, consistent audio** — music and sound design should be gentle and predictable
3. **No cliffhangers** — episodes must have satisfying natural endings, preventing continuous watching
4. **Plot-driven** — a clear narrative arc with a beginning, problem, and resolution
5. **Prosocial themes** — characters modelling sharing, empathy, conflict resolution, etc
6. **Age-appropriate vocabulary** — rich enough to scaffold language development, simple enough to comprehend without frustration
7. **Emotional depth** — characters experience and process emotions visibly, supporting emotion understanding development
8. **Direct address moments** — characters occasionally speak to the child, ask, or pause for response, encouraging active engagement *(optional).*



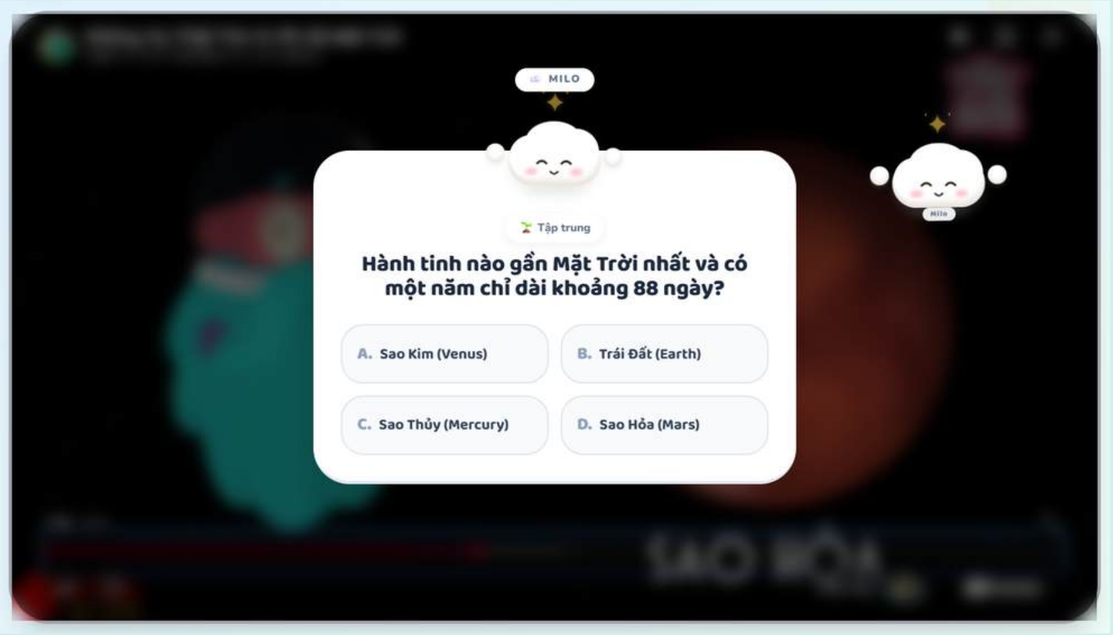
SECTION:

STUDY DESK

Unline traditional studying, children can **watch videos/read books** across **different topics** (e.g. space, nature, ocean, other countries), which can **nurture their curiosity**, encouraging finding out about new knowledge.



While watching, **automatically AI-generated questionares** will be asked by Kindy-Mate about aforementioned information (MCQ, Voice recognition, typing, etc), **encouraging active engagement** instead of passive viewing:



SECTION:
STUDY DESK



AI integration: Kindy-Mate will automatically transcribe the video before drafting questions accordingly.

Transcript: "The Tiny Planet Closest to the Sun"

Host: Today's episode is all about Mercury — the smallest planet in our solar system and one of the hardest to explore.

Guest: Mercury is fascinating because, despite being closest to the Sun, it isn't the hottest planet. That title actually belongs to Venus.

Host: A lot of people get that wrong. Why is Mercury cooler?

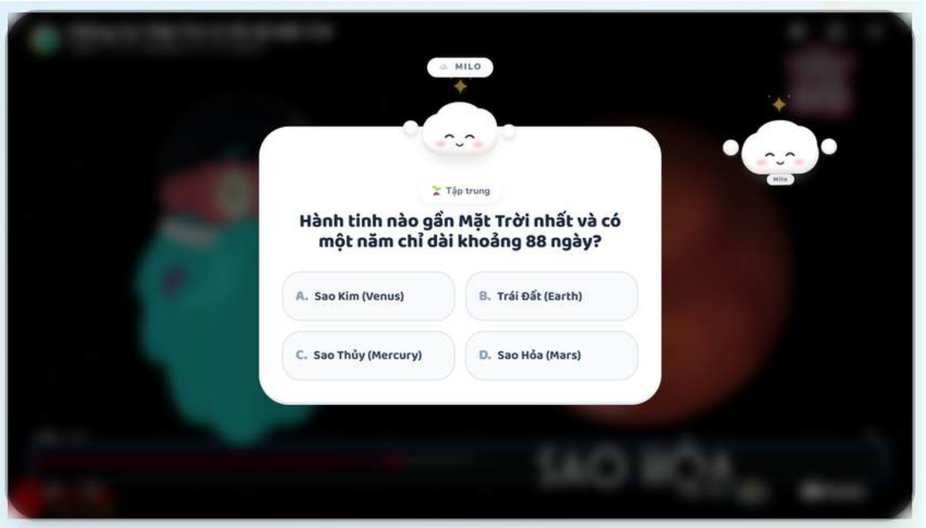
Guest: Mercury has almost no atmosphere to trap heat. During the day, temperatures become incredibly hot, but at night they drop dramatically because the heat escapes into space.

Host: So the temperature difference is extreme?

Guest: Very extreme.

$$430^\circ\text{C} - (-180^\circ\text{C}) \approx 610^\circ\text{C}$$

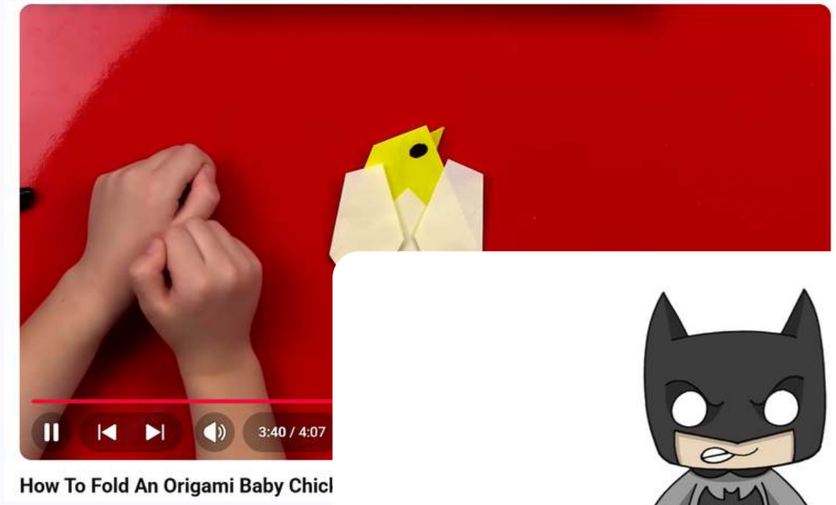
Host: That's a bigger temperature swing than almost anywhere else in the solar system.



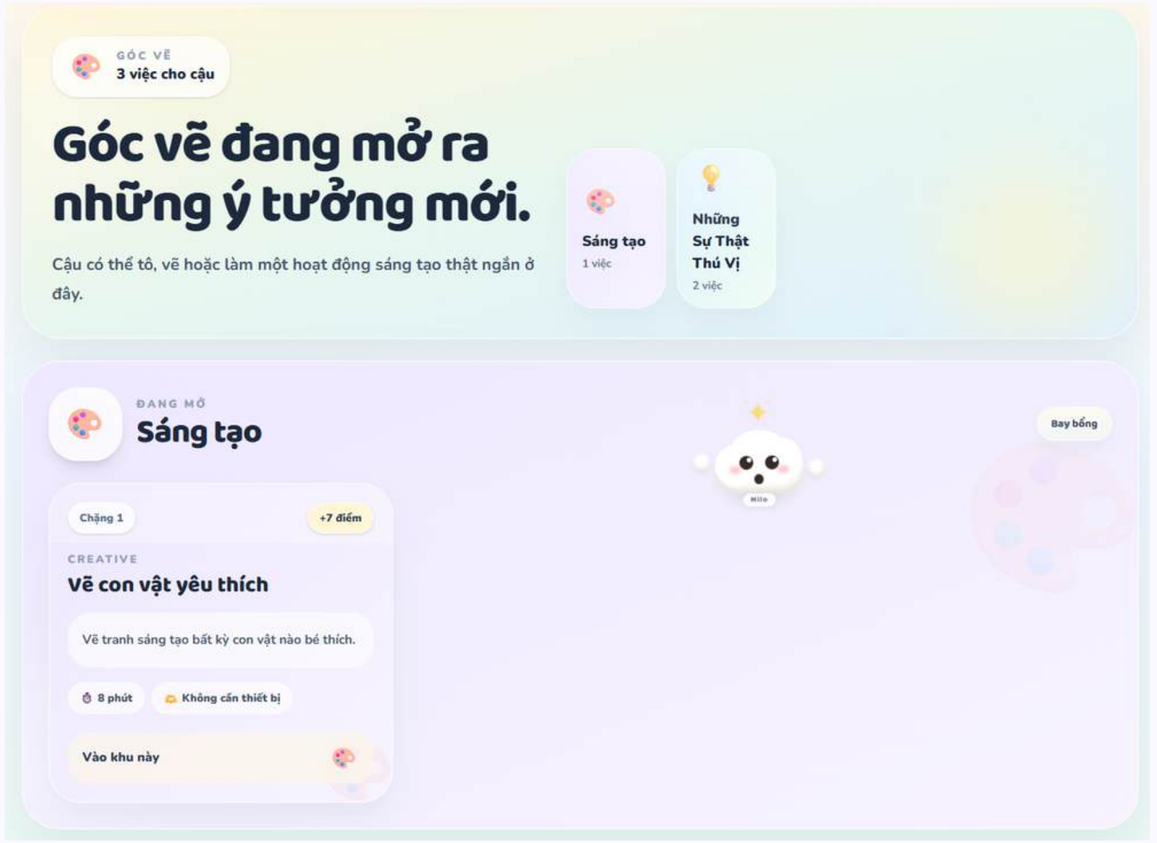
SECTION:

CREATIVE PLAY ZONE

Moving on to real-life activities, children can find **step-by-step video tutorials** guiding children through hands-on creative activities — drawing, colouring, origami, and more. The screen acts as a guide; the experience happens in the real world, done to reduce continuous periods of watching among children.





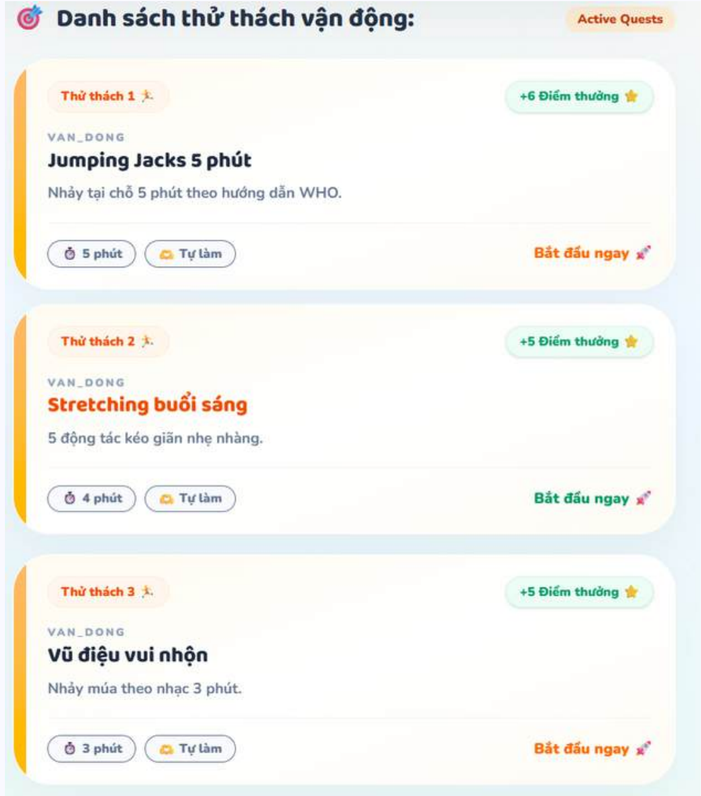


SECTION:

MOVE & PLAY

Short guided video sessions that get children off the screen and work their bodies — through movement **challenges, dance, and active physical activities** they can do at home, giving children a break after a period of screen time.



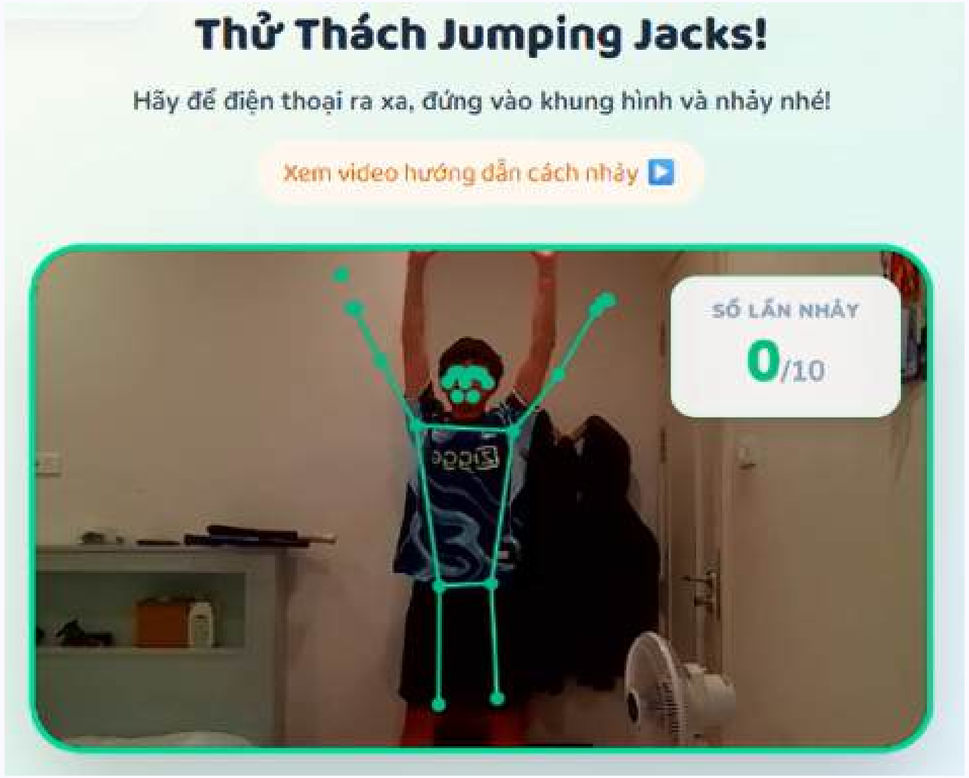


SECTION:

MOVE & PLAY



AI integration: Kindy-Mate will use the camera to track children's movement to make sure they follow and finish the exercise.







SECTION:

GAME ZONE

Children follow characters through adventures and missions. Progression depends on completing quizzes, puzzles, creative challenges, and problem-solving tasks. Learning becomes part of the story experience.

3-5 year-olds

- Simple storytelling and visual interaction
- Short learning activities: colors, shapes, emotions, vocabulary.
- Playful and emotionally engaging design

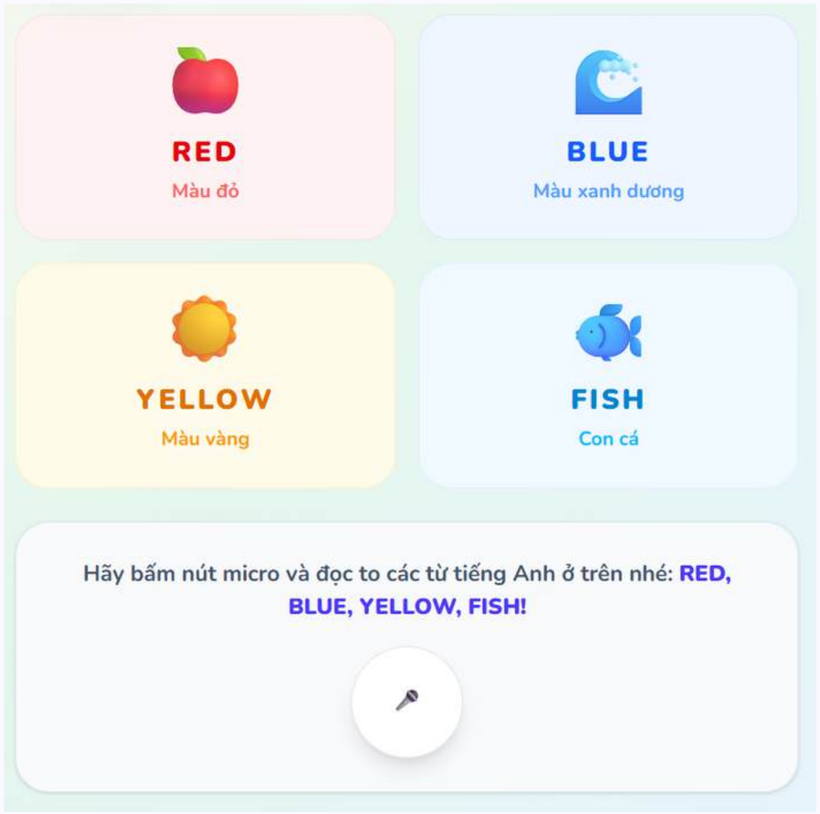
6-8 year-olds

- More complex adventures and missions
- Educational puzzles and mini mysteries
- Activities connected to: language, math, science, and creativity.

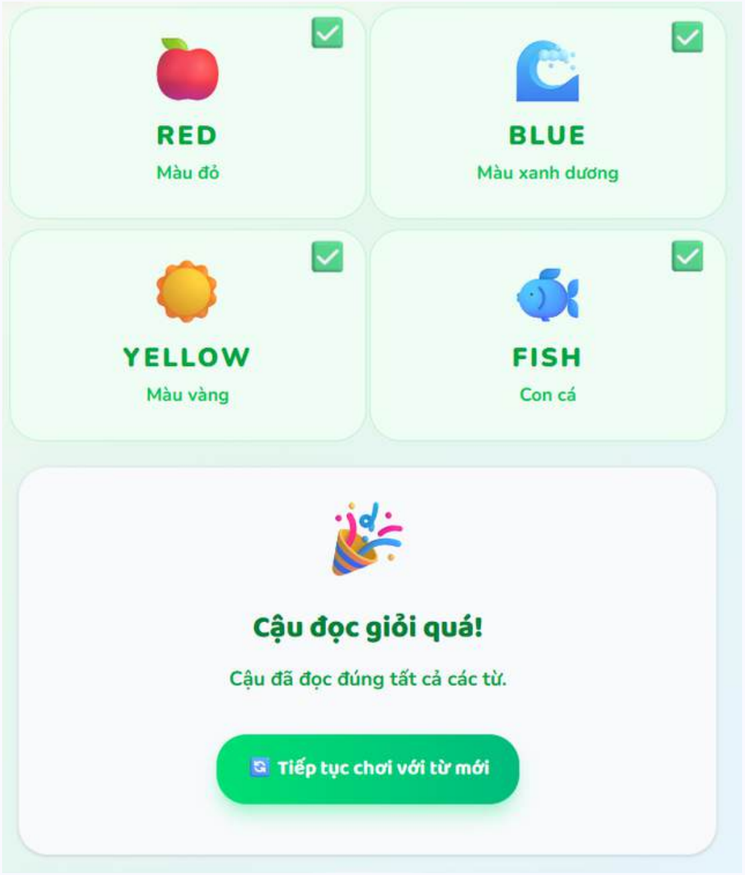
SECTION:
GAME ZONE



AI integration: for Voice recognition games, Kindy-Mate will use the microphone to make sure children read correctly







TIME LIMIT:
FOR EACH AGE GROUP:



Based on the guidelines of UNICEF: **children aged 3-5** should have **less than 1 hour of screen time** per day, while those **aged 6-8** can use **up to 2 hours per day**. Therefore, they are the default time limits our app sets per session for each age group.

Session time adjustment

- Adjust screen-time limits, yet a password is required by parents.
- For excessive increases, the app issues warnings, with the maximum time being ... for each session.
- Parents can customize limits for each section.
- Once a session finishes, reopening for a new session requires a password from parents.

Natural Session Endings

- AI mascot gives a gentle reminder 10 minutes and 5 minutes before time ends:

1. Let's finish this adventure together!🌟
2. You're doing great! We have a little bit of time left 😊

- A timer appears at the 5 minute mark.
- Sections close gradually instead of abruptly.

→ Reduces tantrums and resistance.

If a child watches continuously for 20 minutes, the app will slowly shut down for 20 seconds so children can take a break. The period before each shut down can be adjusted by parents.

session: the maximum time a child can use in one sitting before the app is locked.
In reality, we suspect a child can be allowed to use the screen for more than one session.

A HEALTHY SECTION:

Kindy-Mate avoids either complete freedom or overly rigid control. While young children naturally prefer highly stimulating passive content, fixed activity sequences may reduce motivation

Kindy-Mate:

- Encourages “guided autonomy”: children choose within safe boundaries with structured but flexible experience.
- Always includes at least one offline or low-screen activity, gently encourages activity switching every 15–20 minutes to balance between digital and offline engagement

3-5 year-olds:

- Children can choose what to do first between an on-screen and an off-screen activity.
- Guided transition to movement or creative offline tasks.

→ Reduces continuous screen exposure.

6-8 year-olds:

- Children can choose activity order.
- System nudges to prevent only doing one task and abandoning the plan.

→ Prevents repetitive passive use.
→ Builds self-regulation and healthier digital habits.

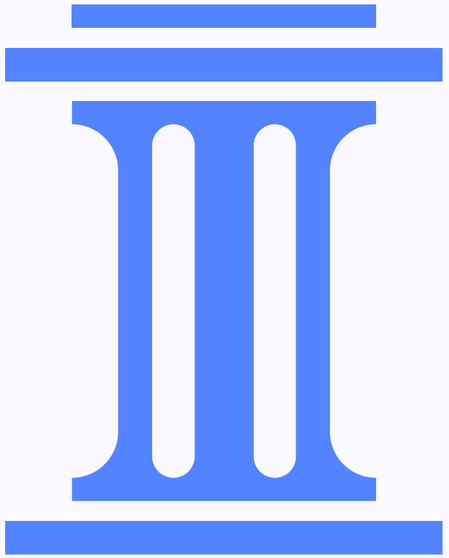
FROM PARENTS' PERSPECTIVE:

HELPING PARENTS SEE, SET, AND SUPPORT HEALTHIER SCREEN HABITS

Parents are able to:

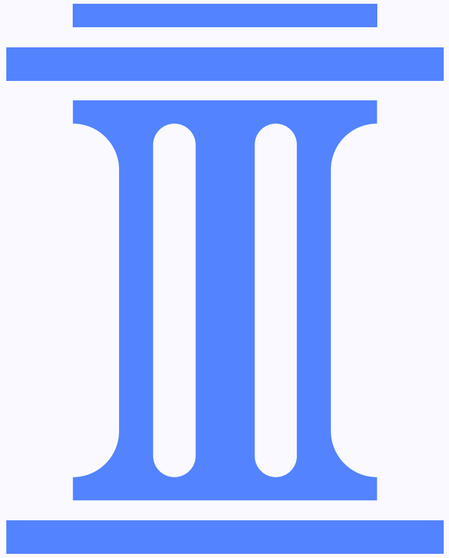
SEE

- Screen-time summary
- Engagement quality
- Activity balance



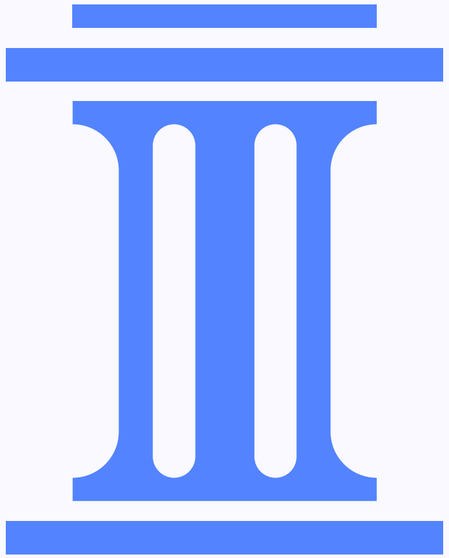
SET

- Time budgets
- Child profiles
- Usage modes
- Permissions



SUPPORT

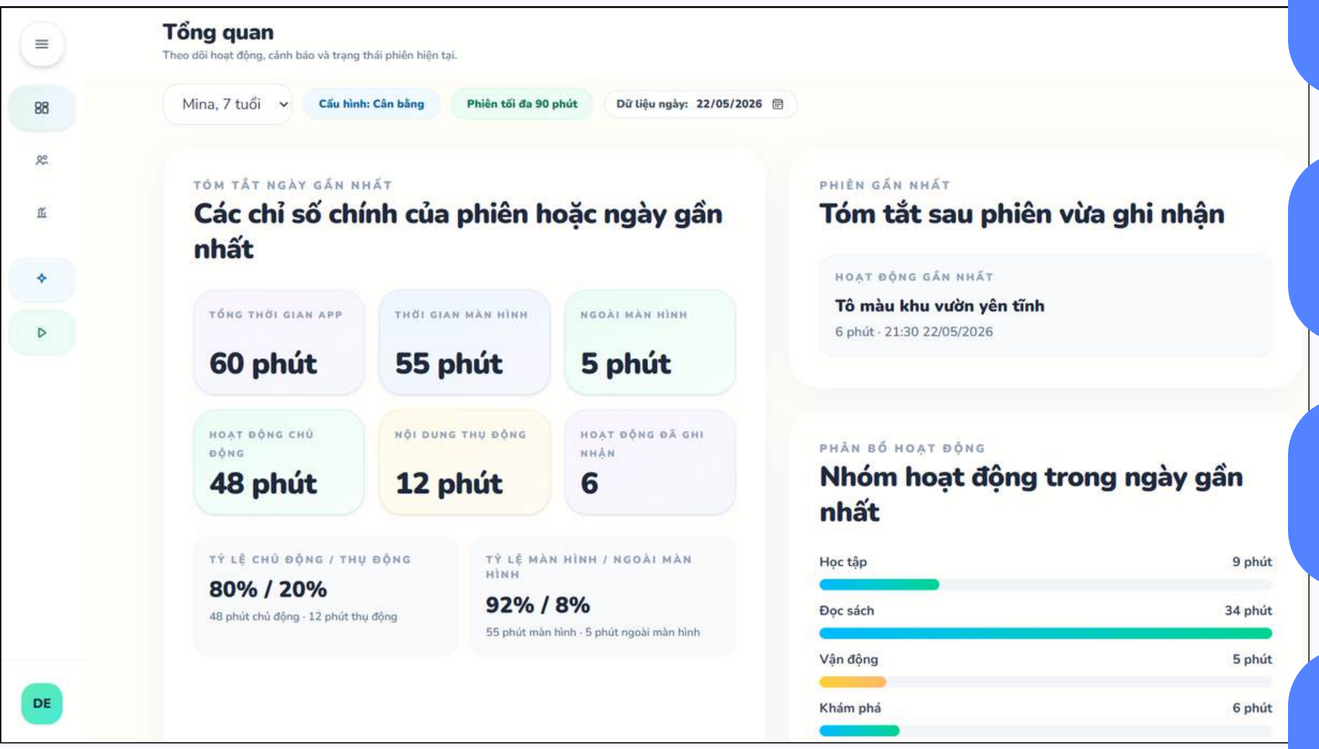
- Behaviour insights
- Contextual recommendations
- Offline activity suggestions



Kindy-Mate gives parents **control without turning screen time** into **daily conflict**.

SECTION 1: SEE THE WHOLE PICTURE

PARENTS CAN UNDERSTAND NOT ONLY HOW LONG CHILDREN USE SCREENS, BUT HOW HEALTHY THAT USAGE IS.



Recent Session Summary — Total screen time, on-screen time, off-screen time

Engagement Quality — Active vs passive engagement

Activity Balance — Learning, reading, movement, exploration

Weekly Pattern — Detects whether usage is balanced or overly passive

SECTION 2: SET GOOD BOUNDARIES

PARENTS SET THE RULES; CHILDREN ENJOY THE EXPERIENCE INSIDE A SAFE STRUCTURE.

Child Profiles

Separate settings for each child

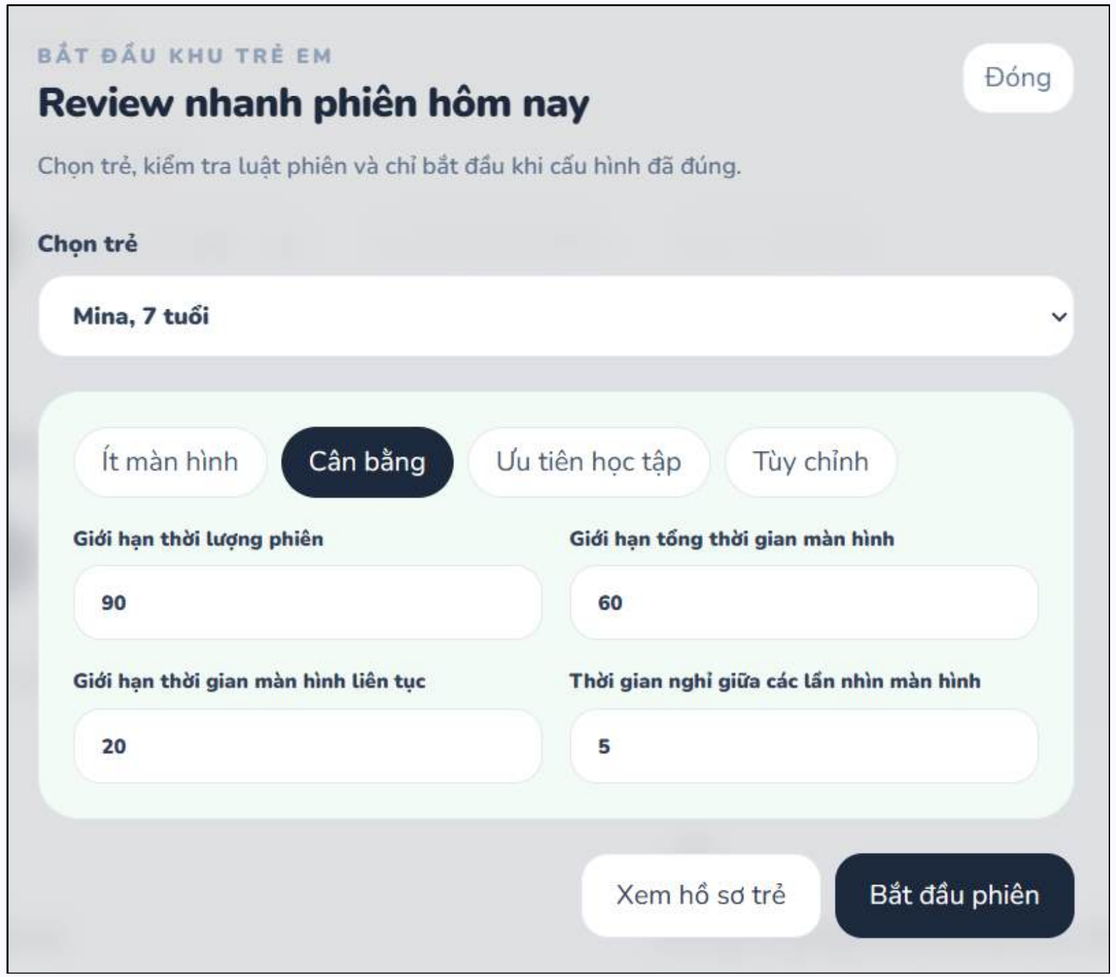


SECTION 2: SET GOOD BOUNDARIES

PARENTS SET THE RULES; CHILDREN ENJOY THE EXPERIENCE INSIDE A SAFE STRUCTURE.

Usage Modes

Low Screen Time / Balanced / Learning-Focused / Custom



Time Budgets

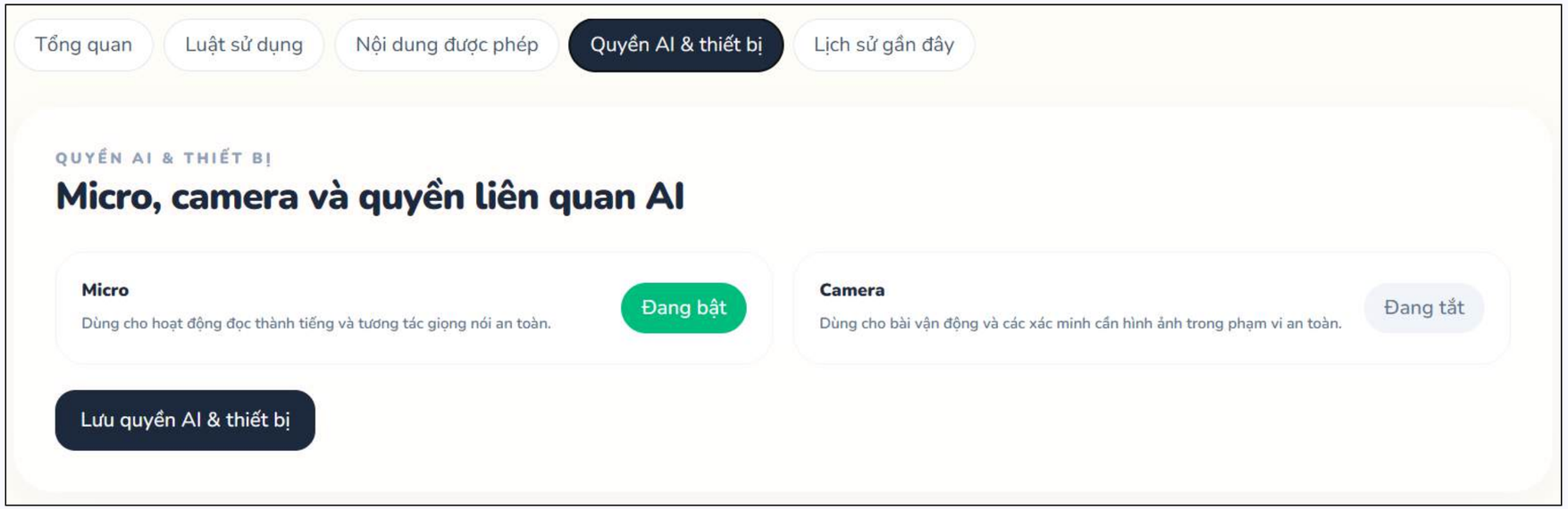
Daily limits and activity-based limits

SECTION 2: SET GOOD BOUNDARIES

PARENTS SET THE RULES; CHILDREN ENJOY THE EXPERIENCE INSIDE A SAFE STRUCTURE.

Safety Permissions

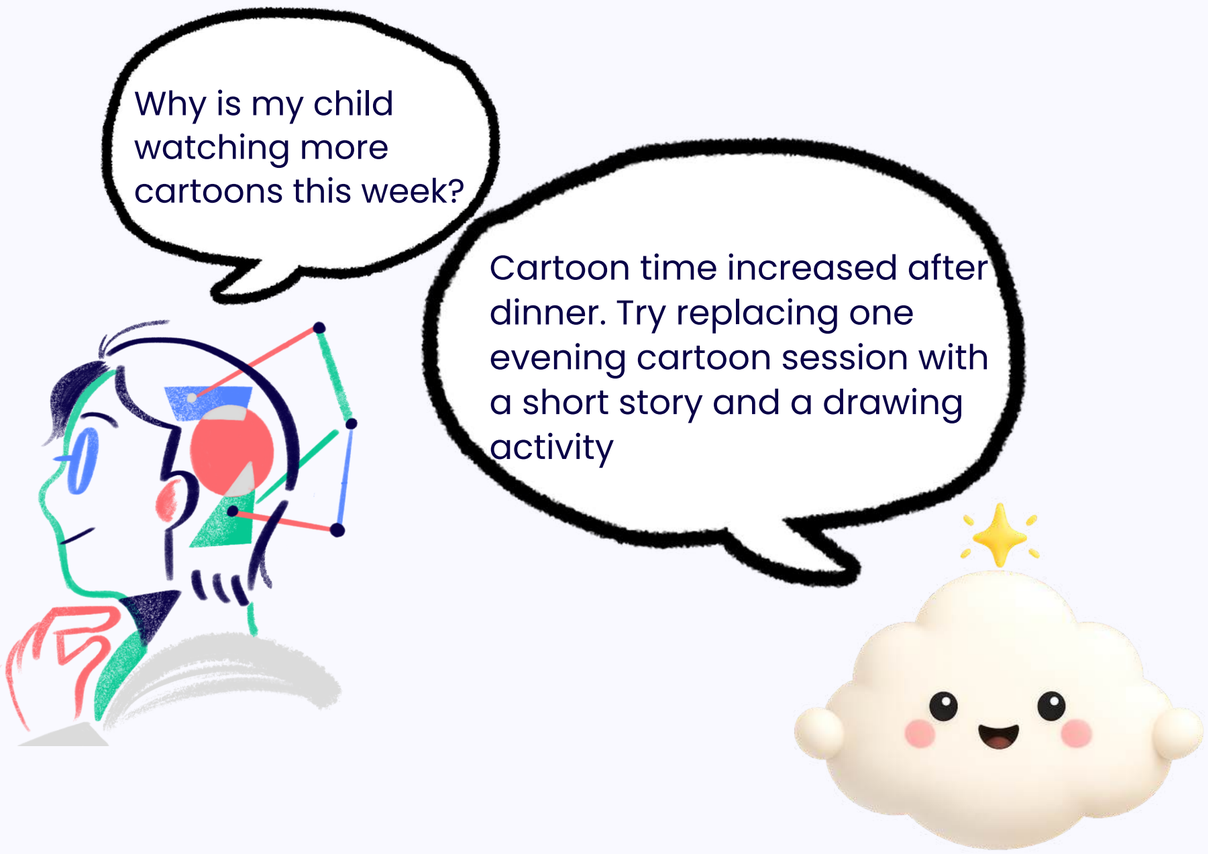
Password, biometrics, AI, camera, and microphone controls



Children cannot modify limits or permissions, reducing common parental-control loopholes.

SECTION 3: SUPPORTED WITH AI

AI TURNS USAGE DATA INTO PRACTICAL RECOMMENDATIONS FOR CHILDREN'S HEALTHIER DIGITAL HABITS.



Analyse

Detects screen-time patterns and engagement trends

Explain

Turns charts into simple parent-friendly insights

Recommend

Suggests offline play, books, toys, or schedule changes.

Children cannot modify limits or permissions, reducing common parental-control loopholes.

SECTION 3: SUPPORTED WITH AI

EXAMPLES OF AI-USE FOR PARENTS

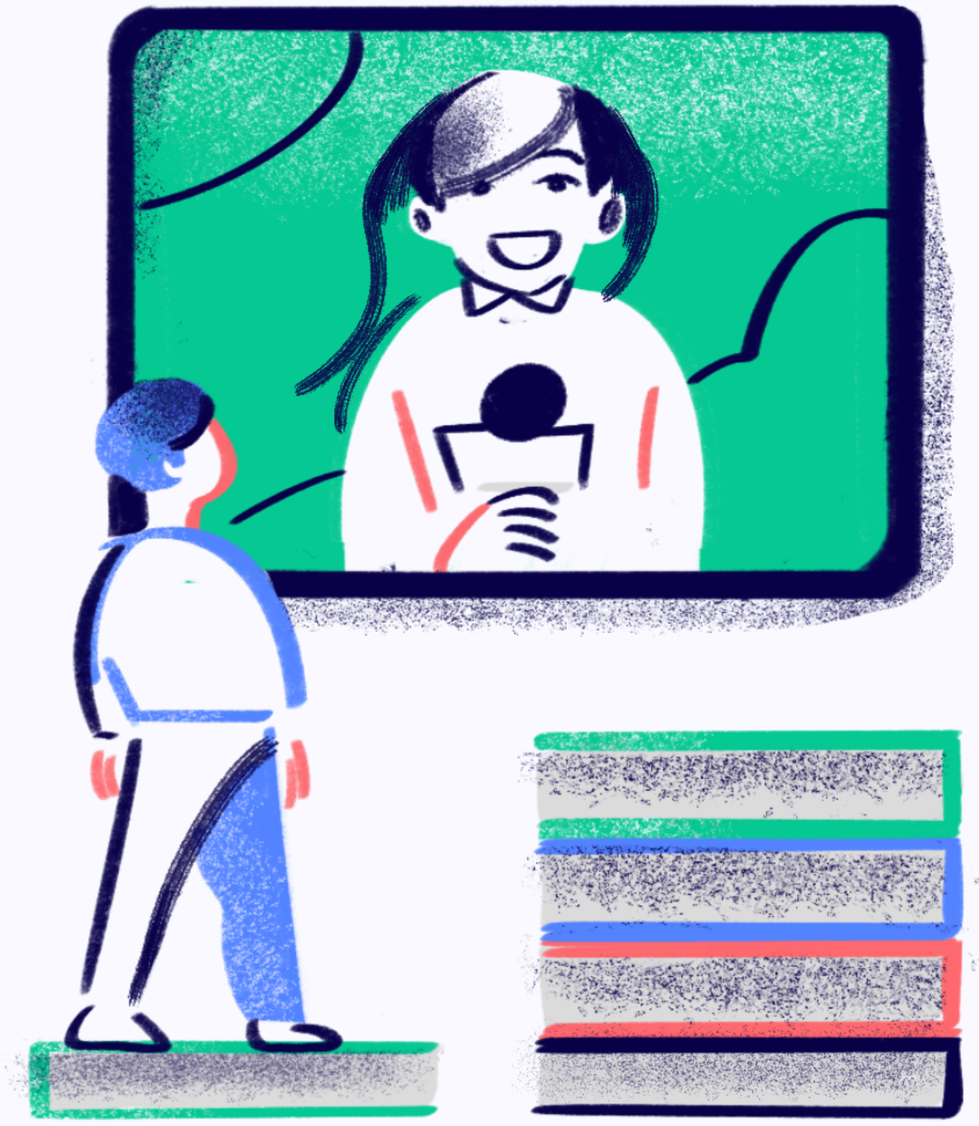


*Parents can chat with AI and ask questions based on their child's recent activity data.

*AI summarizes usage insights and highlights behaviour patterns for parents.



IMPACT, FEASIBILITY, AND SUSTAINABILITY



HOW WE WILL MEASURE SUCCESS

IMPACT IS NOT MEASURED BY LONGER SCREEN TIME, BUT BY HEALTHIER SCREEN-TIME BEHAVIOR.

Child-level Impact

Healthier **cognitive** & **digital habits**

- Learning engagement
- Self-regulated screen behavior
- Offline activity participation
- Reduced reliance on passive entertainment

Parent-level Impact

More **confident** and **informed parenting**

- Clearer view of child screen behavior
- Reduced parent-child conflict
- Data-informed screen-time decisions
- More guided parent-child activities

System-level Impact

Scalable healthy **digital ecosystem**

- Curated healthy content items
- Children reached in pilot
- Schools and parent communities onboarded
- Accessibility across income groups

Our impact is **not measured** by **longer screen time**, but by **healthier screen-time behavior**.

MVP FEASIBILITY & ROADMAP

A WORKING DEMO TODAY, A PILOT-READY PRODUCT IN TWO MONTHS

What we have NOW

- ✓ Demo web with parent and child experiences
- ✓ Child mode with mascot, activity zones, zones' contents
- ✓ Parent dashboard with reports and activity insights
- ✓ Screen-time and off-screen activity tracking foundation
- ✓ and more that we've shown

What we WILL have

- ○ Polish mobile UX for parent and child screens
- ○ Improve mascot AI across the child journey
- ○ Expand curated healthy content for ages 3–8
- ○ Prepare pilot testing with 10-15 families

Feasibility is **already proven**; we are **ready** to turn the demo into a **pilot-ready product**.

SCALABILITY AND LONG-TERM IMPACT

KINDY-MATE CAN GROW FROM A PARENT TOOL INTO A PUBLIC-HEALTH AND EARLY-EDUCATION SUPPORT SYSTEM.

SCALE PATHWAY

A family → A community → A city → Regional impact

SCALING DIMENSIONS

Criteria	Start with...	Grow into...
Content	Vietnamese curated content	Multilingual/localized content
User scale	Individual parent users	Kindergartens, schools, community programs
AI capability	AI question generator, content classifier, and session planner	An adaptive personalized learning journey per child
Impact	Individual family habits	Societal digital parenting norms
Partnerships	Direct-to-parent	Pediatricians, educators, child psychologists, NGOs

SDG ALIGNMENT

- SDG 3 — Good Health and Well-being
- SDG 4 — Quality Education
- SDG 10 — Reduced Inequalities

TIMELINE ROADMAP

after the competition

0–3 MONTHS

Prototype
Content criteria
Parent interviews

3–6 MONTHS

MVP with curated content
AI Q&A Dashboard

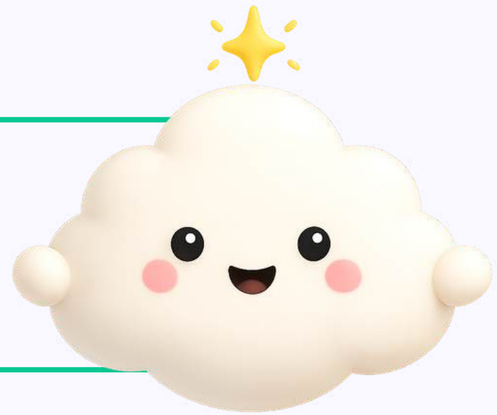
6–12 MONTHS

Pilot with parents / kindergarten / community
12+ months

12+ MONTHS

Scale content library, partnerships, localization

“Kindy-Mate can move from concept to controlled pilot through a staged, safety-first roadmap”



RESPONSIBLE AI AS THE FOUNDATION:

RISK	CRITICAL CONCERN	MITIGATION
AI Hallucination	AI generates inaccurate, age-inappropriate, or confusing responses to children	• Use template-based, closed AI interaction • Pre-approved, human-reviewed content base
Child Data Privacy	Sensitive behavioral and voice data from children collected without adequate protection	• Minimal data collection • Verifiable parental consent before any data use • No third-party data sharing • Full compliance with COPPA 2025 standards
Overuse	High-quality content still causes harm if total screen time is uncontrolled	• Hard parent-set time limits • Every session ends with an offline mission (move, draw, speak, explore)
Inequality of Access	App becomes accessible only to higher-income, tech-literate families -> reinforcing inequality	• Freemium model with core safety features free • Offline-capable content for low-connectivity environments • School/community partnership plan for broader access

THANK YOU



